

SC7 Bayes Methods MT25

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Lecture 2: a first case study in Bayesian inference

Case study: Radiocarbon dating

The data come from the site of an ancient settlement in NZ.

Archaeologists dug down through a habitation band. Above and below this layer there is no sign of activity. They dated $n = 7$ charcoal samples from the habitation layer.

The researchers are very confident (before seeing the data) that

- there was no settlement before $U = 1000$ years BP;
- the settlement had been abandoned prior to $L = 500$ BP.

Key scientific question: For how long was the site settled?

There is a suggestion that this camp was settled for just weeks or months rather than years.

This is a question about the occupation span.

Observation model

Uncalibrated radiocarbon date y_i , $i = 1, 2, \dots, n$ is a biased measurement of true age θ_i of specimen i . Observation model:

$$y_i = \mu(\theta_i) + \epsilon_i$$

with

$$\epsilon_i \sim N(0, \sigma_c(\theta_i)^2 + \sigma_i^2).$$

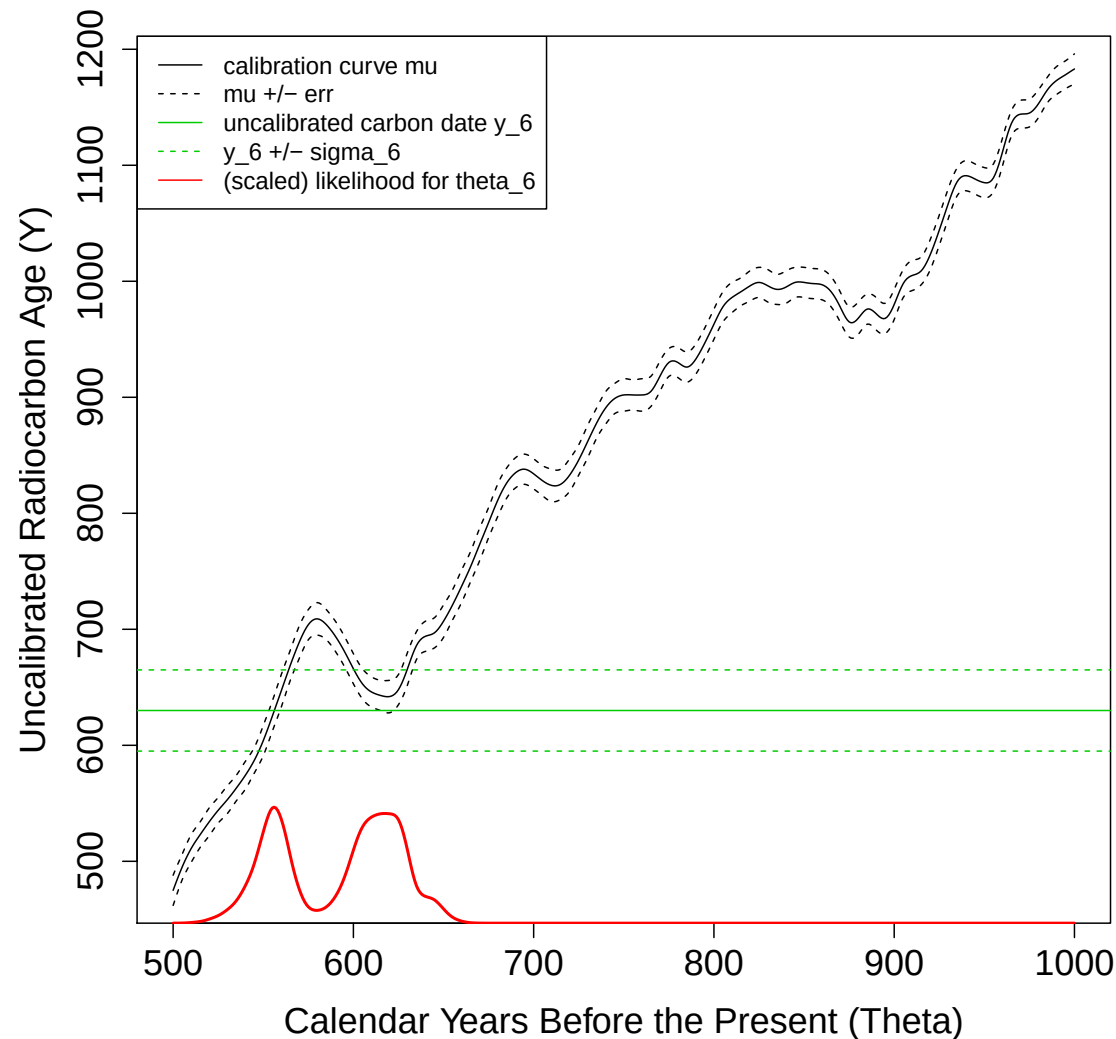
Two sources of noise: measurement error σ_i and uncertainty $\sigma_c(t)$ in $\mu(t)$. Calibration functions $\mu(t)$ and $\sigma_c(t)$ are known.

The likelihood factor for one unknown age parameter θ_i is:

$$p(y_i|\theta_i) = \frac{\exp(-(y_i - \mu(\theta_i))^2 / 2(\sigma_c(\theta_i)^2 + \sigma_i^2))}{\sqrt{2\pi(\sigma_c(\theta_i)^2 + \sigma_i^2)}}$$

Calibration curve μ with likelihood $p(y_6|\theta_6)$ for sixth parameter.

The functions $\mu(\theta)$ and $\sigma_c(\theta)$ are available from radiocarbon dating labs (Eg OxCal, a widely used Bayesian package for RCD).



The data

$n = 7$ radiocarbon dates.

```
#Date id - i - y_i - sigma_i
#NZ 7758 - 1 - 580 - 47
#NZ 7761 - 2 - 600 - 50
#NZ 7757 - 3 - 537 - 44
#NZ 7756 - 4 - 670 - 47
#NZ 7755 - 5 - 646 - 47
#WK 2589 - 6 - 630 - 35
#NZ 7771 - 7 - 660 - 46
```

The likelihood plot in the previous graph showed $p(y_6|\theta_6)$, where the overall likelihood is

$$p(y|\theta) = \prod_{i=1}^n p(y_i|\theta_i).$$

Priors

A simple uniform prior $\pi_u(\theta)$ for $\theta = (\theta_1, \dots, \theta_n)$ might set lower and upper bounds $L = 500$ and $U = 1000$ on the true ages of the specimen, and assert that all other values are equally probable:

$$\pi_u(\theta) \propto \prod_i \mathbb{I}(L \leq \theta_i \leq U).$$

The parameters space is $\Omega_u = [L, U]^n$.

We could estimate the start and end of settlement using $\theta_{(1)} = \min(\theta)$ and $\theta_{(n)} = \max(\theta)$ the oldest and most recent dates, and estimate the span using

$$S_u = \max(\theta) - \min(\theta),$$

.

Does this prior meet our elicitation criteria?

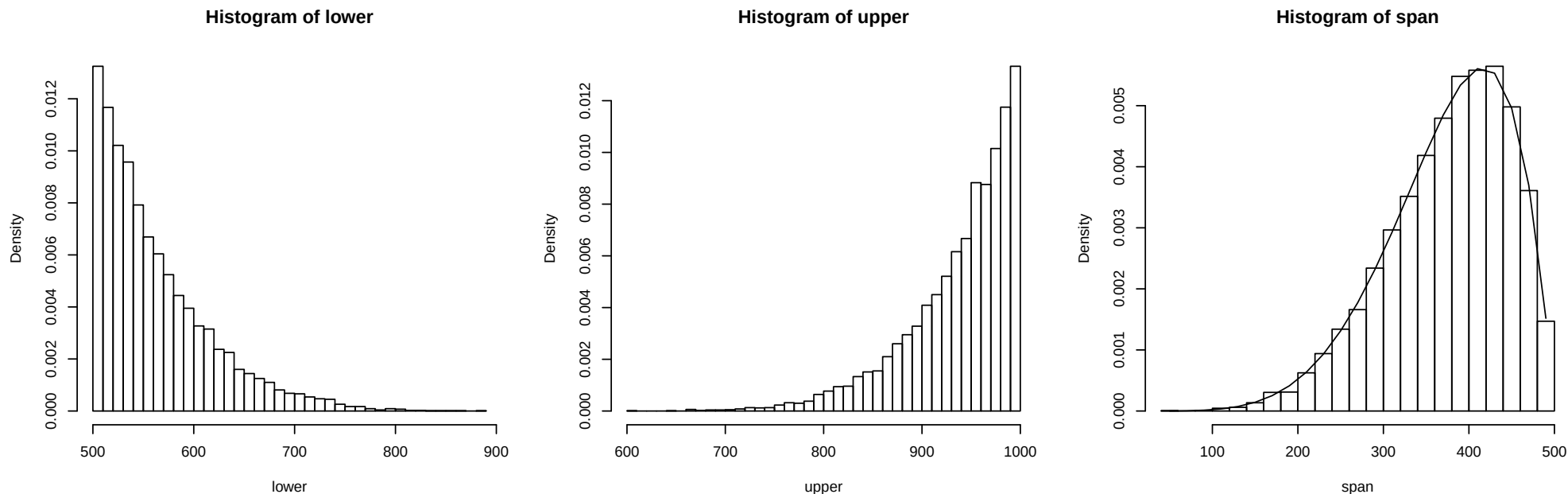
(A) can we think of the dates $\theta_i, i = 1, 2, \dots, n$ as generated by some “physical” process?

(B) the span is linked to a key scientific question - is the prior non-informative on this variable?

Does π_u meet these criteria? Certainly not (A). For (B) we need the marginal prior distribution $\pi_{S_u}(s)$ of the span S_u . This is

$$\pi_{S_u}(s) = \frac{n(n-1)}{(U-L)^n} s^{n-2} (U-L-s) \quad \text{for } 0 \leq s \leq U-L.$$

We don't actually *need* to calculate this as we can get the histogram by simulating the prior.



Marginal priors $\pi_u(\theta^-)$ (left), $\pi_u(\theta^+)$ (mid), $\pi_{S_u}(s)$ (right). We have no basis for favoring large span s values over smaller values.

When $s \ll U - L$ then $\pi_{S_u}(2s) \simeq \pi_{S_u}(s)2^{n-2}$ a strong bias for larger span. We are biasing the thing we want to estimate!

π_u was a bad choice as it does not represent prior knowledge.

A prior from a process generating θ

Model assumptions:

- Settlement starts at time $\psi_2 < U$ and ends at $L < \psi_1 < \psi_2$.
- The span is $S_s = \psi_2 - \psi_1$.
- Dates θ are a realisation of a Poisson process* with rate λ . We choose the number of dates so we condition on n .

Consequences:

- We have $\theta_i \sim U(\psi_1, \psi_2)$ (conditioning a Poisson process on the number of events gives uniform event times).
- Our prior for $\theta|\psi$ is

$$\pi_s(\theta|\psi) = \frac{1}{(\psi_2 - \psi_1)^n} \mathbb{I}(L < \theta_1, \dots, \theta_n < U).$$

*See notes Chapter 1 Appendix B for some background on the Poisson process

We have to specify the prior for $\psi = (\psi_1, \psi_2)$. We would like the span to be uniform. The prior

$$\pi_s(\psi) \propto \frac{1}{U - L - (\psi_2 - \psi_1)}$$

has a uniform distribution[†] on values of $\psi_2 - \psi_1$ so this is non-informative with respect to the span.

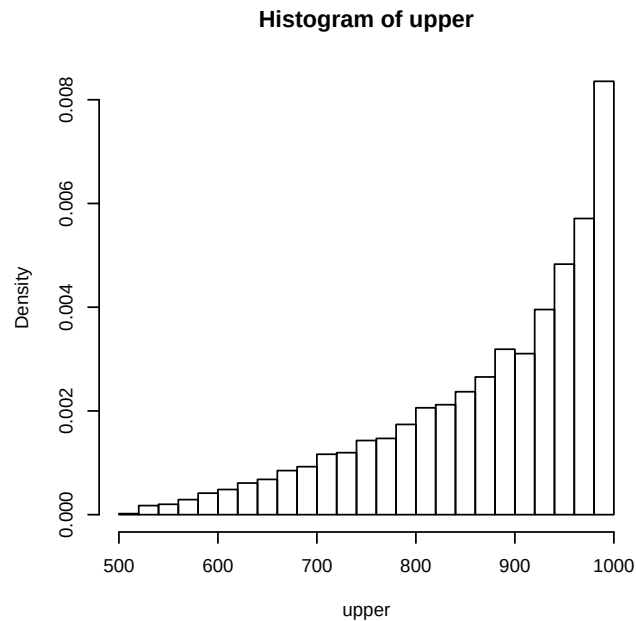
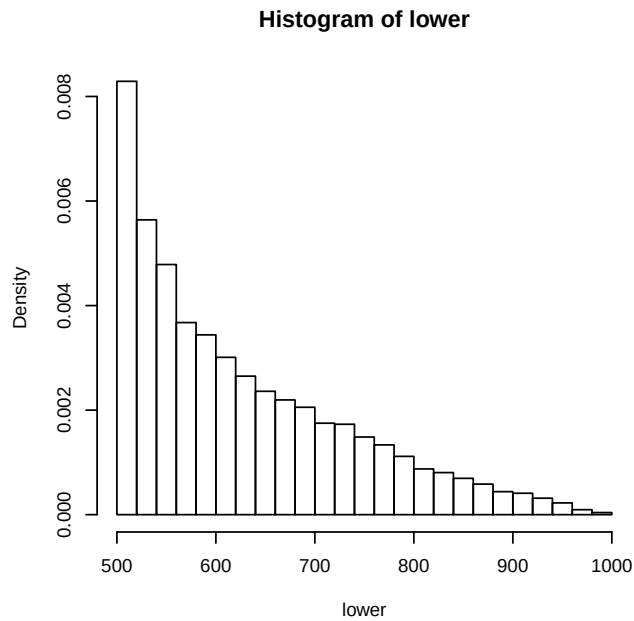
Since $\pi_s(\theta, \psi) = \pi_s(\theta|\psi)\pi_s(\psi)$,

$$\pi_s(\theta, \psi) \propto \frac{1}{(\psi_2 - \psi_1)^n} \frac{1}{(U - L - (\psi_2 - \psi_1))}$$

models the prior information we actually have. The parameter space is

$$\Omega_s = \{(\theta, \psi) \in [L, U]^{n+2} : \psi_1 < \theta_i < \psi_2, i = 1, \dots, n\}$$

[†]See notes Chapter 1 Appendix A for proof



These marginal priors $\pi_s(\psi_1)$, $\pi_s(\psi_2)$ and $\pi_{S_s}(s)$ better represent the prior information we actually possess.

The prior on span is uniform, and the priors $\pi_s(\psi_1)$, $\pi_s(\psi_2)$ distribute probability mass more evenly over the parameter domain.

Radiocarbon example continued ... Sampling the posterior

We have two posterior distributions,

$$\pi_u(\theta|y) \propto p(y|\theta)\pi_u(\theta)$$

and

$$\pi_s(\theta, \psi|y) \propto p(y|\theta)\pi_s(\theta, \psi).$$

The two posterior distributions have the same likelihood, but different priors π_u and π_s , and parameters spaces Ω_u and Ω_s .

We would like to summarise these distributions, form histograms and HPD credible sets.

I used simple random walk Metropolis Hastings MCMC* updating one variable at a time. We look at MCMC later in the course.

I sampled $\theta^{(t)} \sim \pi_u(\theta|y), t = 1, \dots, T$, computed the order statistics $\theta_{(1)}^{(t)}, \theta_{(n)}^{(t)}$ and span $S_u^{(t)} = \theta_{(n)}^{(t)} - \theta_{(1)}^{(t)}$ for each MCMC sample.

I also sampled $(\theta^{(t)}, \psi^{(t)}) \sim \pi_s(\theta, \psi|y), t = 1, \dots, T$, in the second model, and computed the span $S_s^{(t)} = \psi_2^{(t)} - \psi_1^{(t)}$ for each MCMC sample.

I plotted histograms of the samples $S_u^{(t)}$ and $S_s^{(t)}, t = 1, \dots, T$ and used them to estimate HPD sets.

*See attached R-code for details.

Summarising the results

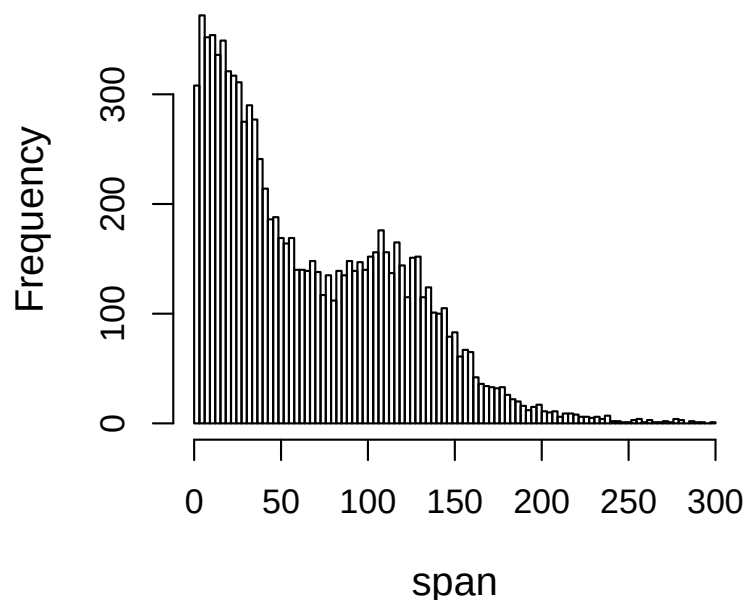
We are interested in the span, and comparing two models with and without shrinkage. Here are the two HPD sets.

Model 2 (shrink/ π_s): [0,160]

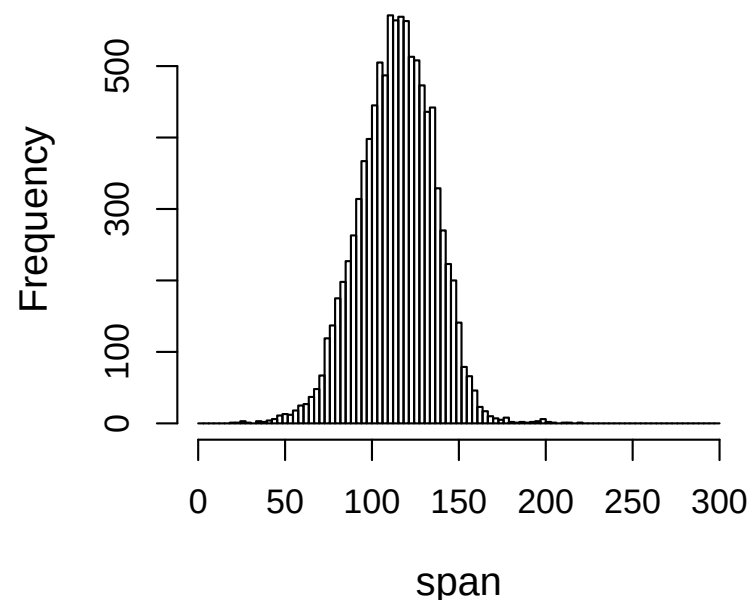
Model 1 (unif/ π_u): [70,160]

The marginal posterior histograms differ in shape and support.

Posterior span, shrinkage prior



Posterior span, uniform prior



Bayes Factor compares models. Which model fits the data best?

Under model one the marginal likelihood $p_u(y)$ say, is

$$p_u(y) = \int_{\Omega_u} p(y|\theta)\pi_u(\theta)d\theta$$

and under model two we have

$$p_s(y) = \int_{\Omega_s} p(y|\theta)\pi_s(\theta, \psi)d\psi d\theta$$

We estimate these using Monte Carlo.[†]

We get $\hat{p}_u \simeq 4 \times 10^{-21}$ and $\hat{p}_s \simeq 8 \times 10^{-19}$.

The Bayes factor $p_s(y)/p_u(y)$ is about $\hat{B}_{s,u} \simeq 200$.

The shrinkage prior π_s is clearly favoured over the uniform prior.

[†]We will see how this is done in a later lecture.

Conclusions

We conclude that a brief settlement time is not ruled out (see histogram of S_s) in our favored analysis. It is ruled out by an analysis using a prior which weights against brief settlement times.

We don't have much data (7 noisy numbers) so the conclusions show some sensitivity to the choice of prior. However the prior π_s better represented actual prior knowledge than π_u and was strongly favored by the data ($B_{s,u} \simeq 200$).

The process-model based prior π_s allows very small spans close to zero, while the uniform prior rules them out. This is clearly a case where we don't want the prior to impose structure we can't support on prior grounds.